

TakeHome - Due Nov 5

Answer the problems on separate paper. You do not need to rewrite the problem statements on your answer sheets. Work carefully. Do your own work. **Show all relevant supporting steps!**

1. (15) Discuss the convergence of five of the following series, i.e., determine where the series converge, where the series converge absolutely and where the series converge uniformly.

$$\begin{array}{lll} \text{a.} & \sum_{n=0}^{\infty} \left(\frac{z}{z+1} \right)^n & \text{b.} \quad \sum_{n=1}^{\infty} n^2 z^{n^2} \quad \text{c.} \quad \sum_{n=1}^{\infty} \frac{z^n}{1-z^n} \\ \text{d.} & \sum_{n=1}^{\infty} \frac{1}{z^2 + n^2} & \text{e.} \quad \sum_{n=1}^{\infty} n^{-z} \quad \text{f.} \quad \sum_{n=0}^{\infty} e^{-nz^2} \end{array}$$

2. (9) Suppose that the radius of convergence of $\sum_{n=0}^{\infty} a_n z^n$ is r , where $0 < r < \infty$. Find the radius of convergence of each of the following series:

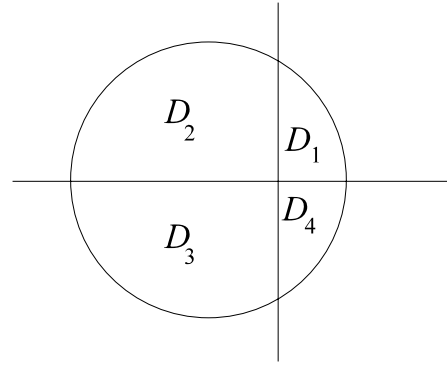
$$\begin{array}{lll} \text{a.} & \sum_{n=0}^{\infty} a_n z^{2n} & \text{b.} \quad \sum_{n=0}^{\infty} a_{2n} z^n \quad \text{c.} \quad \sum_{n=0}^{\infty} a_n^2 z^n \end{array}$$

3. (4) Let $G_1 = \{z : |\operatorname{Im} z| < \frac{\pi}{2}, \operatorname{Re} z > 0\}$. Let $f(z) = \sinh z$. Find the conformal image of G_1 under f . Explicitly show that the images of the curves $m_y = \{z = x + iy : x > 0\}$ and $n_x = \{z = x + iy : -\frac{\pi}{2} < y < \frac{\pi}{2}\}$ under f intersect orthogonally in $f(G_1)$.
4. (4) Let $G_2 = \{z : |z - \frac{1}{2}| < \frac{1}{2}\}$ and $G_3 = \{z : 0 < \operatorname{Im} z < 2\}$. Find a one-to-one conformal mapping f which maps G_2 onto G_3 .

5. (10) The lines, $x = \frac{1}{2}$ and $y = 0$ divide D , (the unit disk centered at 0) into 4 subregions D_1, D_2, D_3 and D_4 .

See figure to the right. Let $w = \frac{z}{z-1}$. Find the images E_j of each subregion D_j under w , i.e., find

$$E_j = w(D_j), \quad j = 1, 2, 3, 4.$$



6. (4) Let $G_4 = \{z = re^{i\theta} : 0 < r < 1, 0 < \theta < \pi/2\}$ and let $G_5 = \{z = re^{i\theta} : 1 < r < \infty, 0 < \theta < \pi/2\}$, i.e., G_4 is the intersection of the first quadrant with the unit disk centered at the origin (the interior thereof) and G_5 is the complement of the closure of G_4 in the first quadrant (the interior thereof). Find a one-to-one conformal mapping f which maps G_4 onto G_5 .
7. (4) Let $u(x, y) = x^3 - 3xy^2 + 2e^y \sin x - 2y + 5$. Show that u is harmonic on RHP (the open right half-plane) and find a harmonic conjugate v such that $f = u + i v$ is analytic on RHP .